

Fixed Income

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1 Market Foundation

1.1 The Global Bond Market

Core Concept 1.1: Market Size & Composition

The fixed-income market represents the largest financial market globally. By the end of 2016, the total value of traded bonds worldwide exceeded \$101 trillion, surpassing both global GDP (\$79 trillion) and total global equity market capitalization (\$65 trillion).

This vast market comprised approximately 2.1 million distinct government and corporate bonds. Corporate bonds accounted for the majority, with roughly 1.6 million outstanding issues, reflecting the immense diversity of corporate debt instruments.

The United States Treasury market, while consisting of only 1,086 actively traded bonds and bills, represented \$14 trillion in value. Furthermore, the US corporate bond sector constituted the second-largest segment at \$8 trillion. The sheer scale and liquidity of the US bond market are primary drivers of the US dollar's central role in global currency markets.

Core Concept 1.2: Historical Development

The modern bond market, much like the foreign exchange market, is a relatively recent development. Its expansion was primarily driven by interest rate liberalization and the dismantling of capital controls following World War II, which greatly facilitated sovereign borrowing. Historically, governments primarily issued debt to finance extraordinary expenses such as wars and the administration of justice. However, the rise of the modern welfare state has fundamentally shifted this dynamic. As developed nations expanded public services, including healthcare, education, and social safety nets, government spending as a proportion of GDP increased significantly, leading to a corresponding structural reliance on debt financing.

Core Concept 1.3: US Treasuries & Foreign Nations

Following the transition to floating exchange rates in 1971, many nations were compelled to accumulate substantial foreign exchange reserves to defend their currency pegs. Additionally, central banks frequently purchase US government bonds, selling their domestic currencies in the process, to deliberately weaken their own exchange rates to support export economies. US Treasuries serve as a globally preferred, secure asset due to several distinct advantages: the unparalleled creditworthiness of the US government, its taxation authority over the world's wealthiest economy, the unique privilege of issuing the primary global reserve currency, and the unmatched liquidity of the US bond market. Consequently, during periods of financial instability, investors instinctively seek refuge in US Treasuries. This surge in demand drives bond prices upward, which concurrently forces bond yields downward, reflecting the fundamental inverse relationship between bond prices and yields.

Core Concept 1.4: Corporate Bond Market

Corporations primarily access the bond market for debt financing due to favorable tax treatments and term structures. In most tax jurisdictions, interest payments on debt are tax-deductible, effectively reducing a firm's taxable income and lowering its overall tax burden. Furthermore, the corporate bond market provides access to long-term financing that is generally unavailable through traditional commercial banks, which tend to prefer issuing shorter-term business loans.

1.2 Bond Mechanics

Core Concept 1.5: Bond Structure

Fundamentally, bonds are promissory notes that obligate the issuer to make regular, fixed interest payments, known as coupons, to the holder. Upon maturation, the issuer is also required to return the initial borrowing amount, referred to as the principal. This predictable stream of cash flows is the defining characteristic of a fixed-income instrument.

Core Concept 1.6: Bond Yields

A bond's yield can be conceptualized as being similar to the annual percentage rate (APR) on a loan or an advertised deposit rate for a savings account. However, while the income generated by a bond is fixed, its yield is not. The primary variable in calculating a bond's yield is its current market price, which fluctuates continuously based on market conditions. Therefore, the yield represents the equivalent interest rate a theoretical bank account would need to offer to produce an identical final return, assuming all coupon payments are deposited immediately upon receipt.

Core Concept 1.7: Price-Yield Relationship

A fundamental principle of fixed-income markets is the inverse relationship between a bond's purchase price and its yield. If an investor pays a lower up-front price for a set of fixed future cash flows, the effective rate of return, or yield, naturally increases. This dynamic is perfectly illustrated by a perpetual bond—an instrument whose principal is never repaid—where price and yield operate as exact mirror images. Consequently, high market yields are advantageous for prospective lenders but detrimental to new borrowers, whereas low yields benefit new borrowers at the expense of lenders. In this context, servicing debt simply refers to the ongoing cost of maintaining existing borrowing obligations through regular coupon payments.

2 Valuation Framework

2.1 Credit Analysis

Core Concept 2.1: Credit Overview

Current bond prices fundamentally reflect market expectations regarding the likelihood of future repayment. However, the standard mathematical calculation of yield assumes that the promised cash flows will materialize completely. Therefore, when market participants begin to doubt a borrower's solvency, they demand a lower initial price for the bond to compensate for the elevated risk. Because the nominal cash flows remain unchanged, this drop in price mathematically results in a sharply higher yield. For sovereign debt, if investors suspect a government may struggle to meet its obligations, the resulting decline in bond prices and subsequent spike in yields significantly increases the cost of future borrowing. Through this mechanism, the bond market actively imposes fiscal discipline on sovereign nations.

Core Concept 2.2: Sovereign Credit Risk Metrics

To assess sovereign creditworthiness, investors primarily evaluate three macroeconomic metrics. First, the Debt-to-GDP ratio serves as a general indicator of leverage; typically, a higher proportional debt burden implies greater default risk, prompting investors to demand elevated yields. However, this is not an absolute

rule, as evidenced by Japan, which maintains exceptionally low borrowing costs despite holding one of the highest debt burdens globally. Second, the Deficit-to-GDP ratio highlights the extent to which a government's expenditures exceed its tax revenues. Persistent deficits require continuous financing through the bond market, subsequently increasing the proportion of the national budget consumed by debt interest payments. Finally, a nation's repayment schedule, or its strategy for rolling over debt, is scrutinized. While issuing short-term debt is attractive due to lower associated interest rates, it exposes the borrower to refinancing risk—the danger that lenders may suddenly demand significantly higher yields. Due to its robust credit profile, the US government is uniquely positioned to exploit the cost advantages of short-term borrowing and continuous refinancing without substantial fear of abrupt lender withdrawal.

Core Concept 2.3: Credit Risk Indicators

Market participants rely on several formal and market-implied indicators to gauge credit risk. Credit rating agencies assign letter grades to bond issues, ranging from AAA down to D. Within the corporate sector, bonds rated BBB- and above are classified as "investment grade," whereas those rated BB+ and below are categorized as "non-investment grade" or colloquially as "junk bonds." It is widely noted that the historical performance of these agencies has been less reliable for corporate issuances compared to sovereign debt. Alternatively, Credit Default Swaps (CDS) function as a form of market-traded insurance against bond defaults. Because they are actively traded instruments, CDS spreads provide a real-time assessment of credit risk that frequently serves as a more timely early warning indicator than static agency ratings. Simply put, a wider CDS spread denotes higher perceived market risk.

2.2 Macroeconomic Drivers

Core Concept 2.4: Macroeconomic Drivers

Beyond creditworthiness, bond valuations are profoundly influenced by macroeconomic factors, specifically short-term interest rates and inflation expectations. Short-term interest rates exert a direct and immediate effect on broader market yields. In the sovereign space, non-US government bonds invariably price themselves relative to US Treasuries. Because the United States is generally perceived as the most creditworthy sovereign borrower, bonds issued by other nations must offer a lower price—and consequently a higher yield—to compensate investors for the additional inherent risk. This dynamic is essentially basic price competition expressed through the terminology of yield. Furthermore, inflation is a critical driver of bond markets. Because bond payments are fixed, inflation steadily erodes the future purchasing power of both coupon and principal disbursements. In response to rising inflation, investors demand lower initial purchase prices for the same nominal cash flows, driving yields upward. It is crucial to recognize that quoted bond yields are nominal figures and do not inherently adjust for inflation. Ultimately, because US government yields serve as the foundational benchmark for global investments, yield movements across international developed markets tend to be highly correlated.

3 Monetary Policy & Market Dynamics

3.1 Central Banking

Core Concept 3.1: Central Bank Mandates

Central banks actively intervene in money markets to steer economies toward stable, sustainable growth, managing two primary psychological extremes: inflation and deflation. Inflation, which often historically spikes during wartime due to excessive monetary expansion, severely erodes the real value of fixed-income assets, as notably witnessed during the economic turbulence of the 1970s. Conversely, deflation is equally destructive, as anticipating consistently falling prices incentivizes consumers to indefinitely delay spending, which stalls economic activity. Central banks must therefore carefully balance these forces, operating with the understanding that unchecked inflation will inevitably drive bond prices down and corresponding yields up.

Core Concept 3.2: Inflation Measures

To assess current inflationary pressures, central banks rely on three primary statistical metrics: the GDP deflator, the Consumer Price Index (CPI), and the Personal Consumption Expenditures (PCE) price index. The PCE is particularly favored by policymakers because it isolates the core inflation trend by stripping out highly volatile food and energy costs. However, because these indices are inherently backward-looking indicators, bond market investors primarily look to specific traded financial instruments to gauge real-time, forward-looking inflation expectations.

Core Concept 3.3: Treasury Inflation Protected Securities (TIPS)

Treasury Inflation-Protected Securities (TIPS) are explicitly designed to shield lenders from purchasing power erosion by adjusting their principal and coupon payments in accordance with changes in the CPI. By analyzing the pricing differential between TIPS and standard, non-protected Treasury bonds of the exact same maturity, market participants can precisely extract the market's implied inflation expectations. When inflation forecasts rise, investor demand for TIPS increases relative to standard bonds, causing the yield spread between the two assets to widen.

Core Concept 3.4: Output Gap

The output gap represents the macroeconomic difference between an economy's actual production and its theoretical potential output. When an economy operates "tightly"—meaning actual output exceeds potential capacity—it frequently signals impending inflation. Conversely, economic "slack" generally coincides with deflationary pressures. Central bankers synthesize data on the output gap alongside traditional inflation metrics to formulate monetary policy, determining whether it is necessary to cool an overheating economy, stimulate a stagnant one, or maintain current interest rates.

Core Concept 3.5: Policy Toolkit

The primary mechanism central banks use to influence the economy is the adjustment of short-term interest rates. By raising these rates, central banks make holding cash deposits more attractive, which systematically dissuades consumer borrowing and corporate investment. This reduction in aggregate demand lowers actual economic output and effectively suppresses inflation. In the United States, the Federal Open Market Committee (FOMC) determines this pivotal rate eight times a year, generating immense market impact as

it directly calibrates US bond yields. Beyond immediate rate changes, the FOMC heavily relies on "forward guidance"—carefully crafted policy statements where minute linguistic shifts signal future interest rate trajectories. Market participants actively price these expectations into the futures and options markets, allowing analysts to derive clear probabilities regarding upcoming monetary policy decisions.

3.2 The Yield Curve

Core Concept 3.6: Yield Curve

The yield curve visually represents the cost of borrowing across various maturation horizons and serves as a fundamental barometer for the global economy. Under normal conditions, the curve slopes upward, reflecting the "term premium"—the additional yield demanded by investors to compensate for the elevated risks and inflation uncertainties inherent in locking up capital for extended periods. The far-left segment of the curve is rigidly tethered to the overnight interest rate established by the central bank. Consequently, when the Federal Reserve adjusts this rate, the effects cascade across the entire \$101 trillion fixed-income market. In contrast, the far-right segment of the curve floats freely, driven predominantly by aggregate market inflation expectations. Because corporate debt and consumer loans are priced as a spread above the sovereign yield curve, its movements dictate global borrowing costs. Furthermore, the curve's shape provides vital economic signaling: a steepening curve typically indicates anticipation of robust economic growth, a flattening curve suggests deteriorating conditions, and an inverted yield curve is historically recognized as a reliable precursor to an economic recession.

Core Concept 3.7: Market Impacts

The dynamics of the government yield curve exert profound impacts across all sectors of the economy. In the corporate sector, firms typically match the duration of their liabilities with their multi-year capital investments by issuing medium to long-term bonds. Because corporations possess inherently lower creditworthiness than sovereign issuers, their debt commands a higher yield. The structural difference between a corporate bond's yield and a maturity-matched government bond is known as the "spread." As central banks manipulate the foundational government yield curve, corporate yields generally move in tandem, maintaining relatively stable spreads. However, fluctuations in the specific "corporate spread"—the yield gap between a 10-year US Treasury and a 10-year investment-grade corporate bond—reflect shifting market risk appetite, with expansions termed "widening" or "steepening" and contractions termed "tightening" or "flattening." These market dynamics directly affect consumers, as key lending instruments like the 30-year US mortgage rate are tightly anchored to 10-year government bond yields, heavily influencing broader housing market activity. Additionally, due to the highly integrated nature of global finance, benchmark 10-year sovereign yields across advanced economies remain strongly correlated, moving synchronously in response to overarching macroeconomic trends.

Core Concept 3.8: Yield Curve Dynamics

The structural dynamics of the US Treasury yield curve are principally driven by short-term interest rates and forward-looking inflation expectations, as sovereign credit risk is effectively considered negligible. This environment allows astute investors to position themselves based on forecasts of the Federal Reserve's policy shifts and broader inflationary trends. The left-hand anchor of the yield curve acts in lockstep with the Federal Funds Target Rate, shifting discretely based on formal FOMC decisions. Conversely, the right-hand tail of the curve is unanchored, freely oscillating according to the natural supply and demand forces exerted by bond investors reacting to ingrained inflation expectations. Therefore, as inflation forecasts shift, the long end of the curve actively re-prices while the short end remains strictly tethered to prevailing central bank policy.

Core Concept 3.9: Economic Signals

The predictive power of the yield curve is well-documented in modern economic history, such as during the recovery phase following the early 2000s dot-com contraction. As economic activity initially accelerated, market participants anticipated that the Federal Reserve would preemptively hike short-term interest rates to curb emerging inflation. Fearing that rising rates would erode bond values, investors aggressively sold off long-dated bonds, driving long-term yields higher and creating a steeply sloping yield curve—a classic hallmark of an accelerating economy. Subsequently, as the central bank actually executed these short-term rate hikes, the economic outlook shifted. Anticipating that higher borrowing costs would inevitably constrain growth and cool the economy, investors rushed back into the safety of long-term bonds to lock in the elevated yields. This surge in demand drove long-term yields back down, ultimately forcing long-term rates below short-term rates. This phenomenon resulted in an inverted yield curve, serving as a quintessential leading indicator for an impending economic downturn.